

# ApexPlanet Data Analytics Internship Final Report

## 1. Executive Summary

This project involved performing end-to-end data analytics on the Superstore dataset. The objective was to transform raw business data into actionable insights using Python, SQL, Power BI, and Machine Learning techniques.

## 2. Project Objectives

- Clean and preprocess business data
- Perform Exploratory Data Analysis
- Extract business insights using SQL
- Build an interactive dashboard
- Apply statistical analysis
- Develop predictive models

## 3. Dataset Overview

Dataset: Superstore Dataset

Records: 9,977

Features:

- Sales
- Profit
- Quantity
- Discount
- Category
- Region
- State
- Segment

## 4. Tasks Completed

### Task 1

Data Cleaning and Exploratory Data Analysis

### Task 2

SQL Data Extraction and Business Analysis

### Task 3

Power BI Dashboard Development

### Task 4

Advanced Analytics and Predictive Modeling

## 5. Key Insights

1. Technology generated the highest sales revenue.
2. West region achieved the highest profit.
3. Consumer segment contributed the largest share of sales.
4. Phones were the highest-selling sub-category.
5. Customer records were successfully segmented using K-Means Clustering.

## 6. Machine Learning Results

Linear Regression Model

MAE: 59.80

RMSE: 250.86

R<sup>2</sup> Score: 0.1741

## 7. Business Recommendations

- Focus marketing efforts on Technology products.
- Expand successful strategies used in the West region.
- Increase retention efforts for high-value customer segments.

## 8. Future Scope

- Implement advanced forecasting models.
- Build real-time dashboards.
- Deploy predictive models using cloud platforms.

## 9. Conclusion

The internship provided practical experience in data cleaning, SQL, dashboard development, statistical analysis, and machine learning. The project successfully transformed raw data into actionable business insights.

## Appendix

Figure 1: Power BI Dashboard

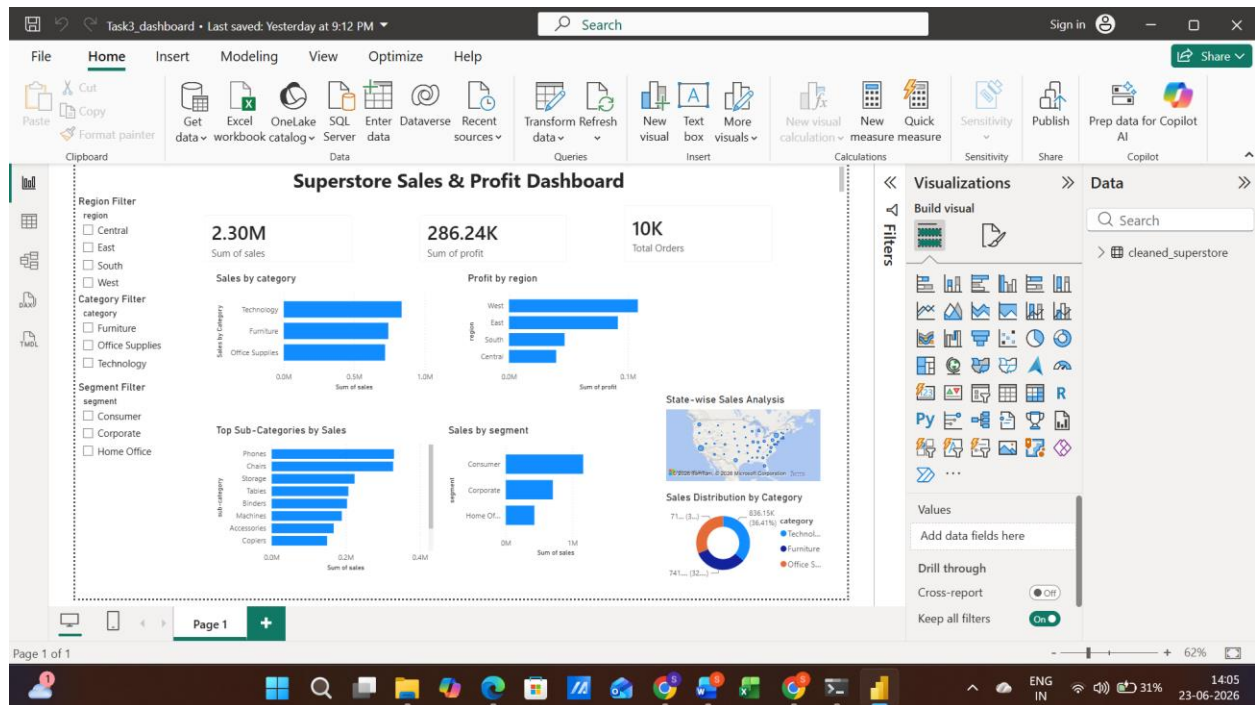


Figure 2: Customer Segmentation using K-Means Clustering

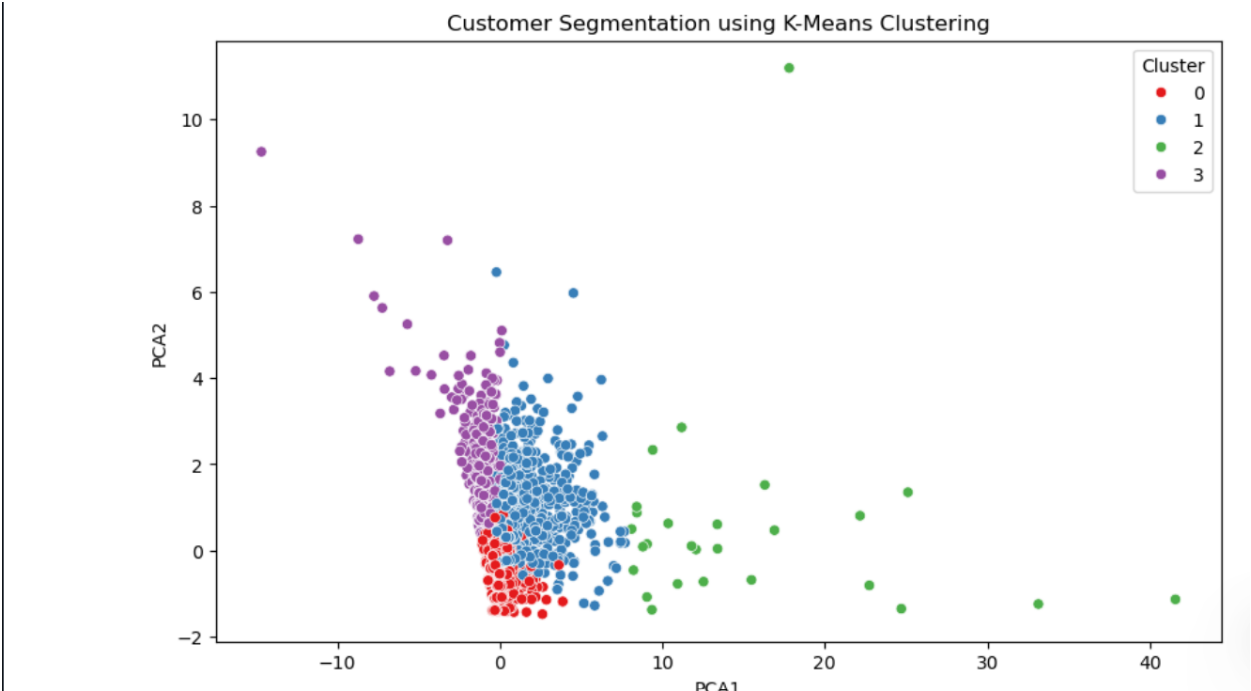


Figure 3: Actual vs Predicted Profit

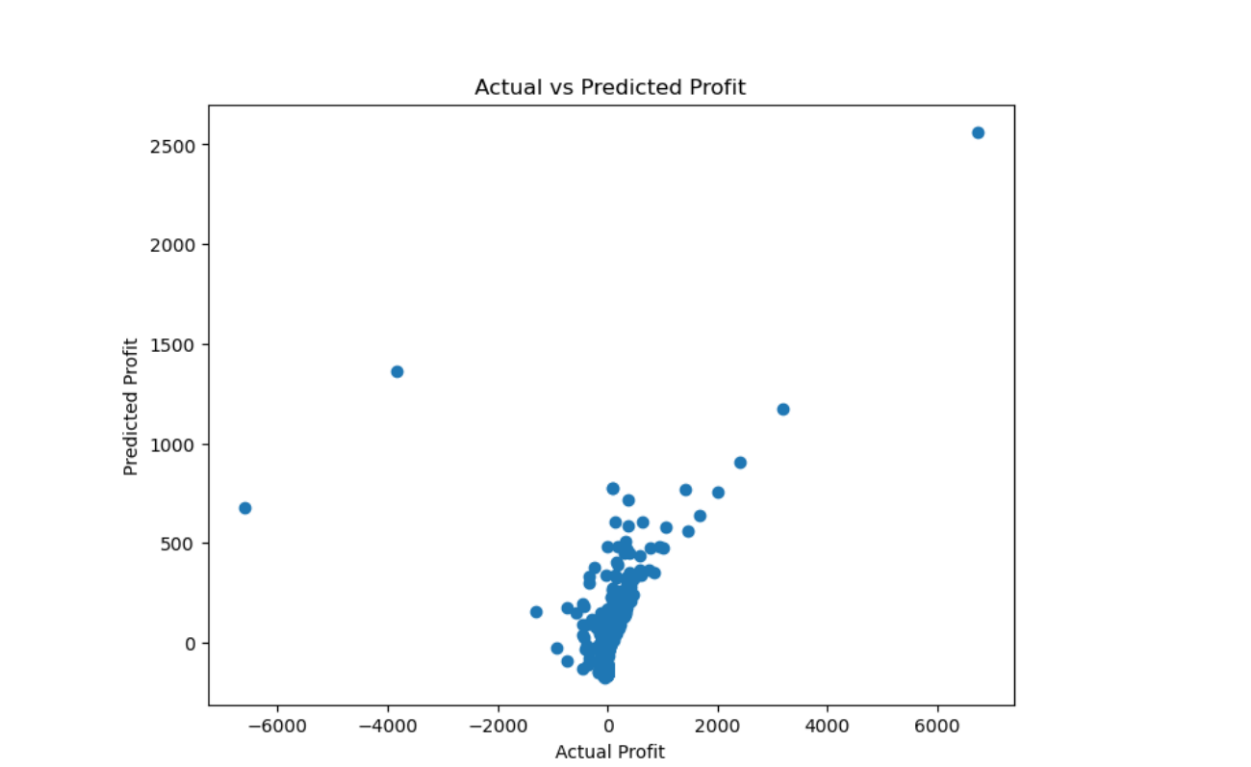


Figure 4: GitHub Repository Structure

- ▼  dashboards
  -  Task3\_dashboard.pbix
- ▼  data
  - >  database
  - >  processed
  - >  raw
- ▼  notebooks
  - >  .ipynb\_checkpoints
  -  EDA\_Task1.ipynb
  -  SQL\_Task2.ipynb
  -  Task4\_Advanced\_Analytics.ipynb
- >  scripts
-  README.md